Grip or glide? The surface challenge · Tier Pack · Aspire

Learning goal: I can explain how friction acts between touching surfaces and judge when grip or glide is useful.

Aspire Route (Book First)

Explain how static grip helps a person walk forwards while kinetic friction slows a sliding box.

Help: Friction opposes relative slipping at the contact: when a foot pushes backwards on the ground, grip prevents backward slip.

Optional Make It · Aspire

Make an engineering trade-off card

Design a tyre tread card comparing icy roads and dry tracks. Explain why one surface pair does not suit every job. Keep your book notes too.

Word Help

friction: a contact force between two surfaces that slide or try to slide past each other

contact force: a force that acts only when two objects or surfaces touch

grip: high resistance to slipping between two touching surfaces

lubrication: adding oil or grease to reduce unwanted friction between moving parts

Exit Check

Which way does friction point when a block slides right?

Name two touching surfaces where high friction is helpful.

What information do you need before deciding if friction is helpful or unhelpful?